

Learning Objective 3.9: Calculate null-steering weights, implement pattern nulls on the ADALM-PHASER, and compare manual null steering with adaptive beamforming.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **From weights to GUI settings.** A null-steering calculation for a 4-element subarray returns the complex weights

$$w_1 = 0.62 - j0.13, \quad w_2 = 0.88 + j0.05, \quad w_3 = 0.88 - j0.05, \quad w_4 = 0.62 + j0.13$$

The Phaser GUI takes element amplitude as a percentage of the largest weight and element phase as an offset in degrees.

- (a) Compute the four magnitudes and convert them to the percentages you would type into Element Gains.

gains =

- (b) Compute the four phase offsets in degrees.

phases =

- (c) A student enters the four gain percentages but forgets to enter the phase offsets. Describe the pattern they will measure and say whether a notch appears.

- (d) All four weights are multiplied by e^{j40° before being entered. How does the measured pattern change, and why?

2. **Reading a null-steered sweep.** A student sweeps the uniform 8-element array, freezes the trace, enters null-steering weights for a null at $+22.5^\circ$, and sweeps again. The two traces read as follows.

	Uniform (frozen)	Null steered
Peak level	0.0 dB (reference)	−1.8 dB
Level at $+22.5^\circ$	−12.8 dBc	−21.6 dBc

- (a) How much additional rejection did the null steering buy at $+22.5^\circ$?

$\Delta =$

- (b) Express the notch depth relative to the null-steered array's own peak, which is the number a jammer-to-signal calculation needs.

depth =

- (c) A second student reports a notch at -31 dBc on the same bench with the same weights. The sweep's noise floor sits about 23 dB below the uniform peak. Is the report credible? Explain.

- (d) The notch appears centred at $+19.7^\circ$ rather than $+22.5^\circ$. Give the most likely cause.

3. **Null depth under coarse quantization.** The lab is repeated on a board whose phase shifters have only $B = 2$ bits.

(a) Find the phase LSB.

LSB =

(b) Use the rule of thumb that quantization sidelobes sit near $-6B$ dB to estimate the deepest notch this board can hold.

depth $\approx$

(c) Round the four phases from Question 1(b) to the nearest multiple of the 90° LSB. What does the resulting weight vector do?

phases =

(d) Explain why holding a deep null is more demanding of phase resolution than holding a beam peak.

4. **Two-channel MVDR by hand.** The PHASER's two digital channels are steered so that a target off boresight presents the real steering vector $s = [1, -1]^T$, while an interferer on boresight presents $u = [1, 1]^T$ with power $P = 9$ relative to a per-channel noise power of 1. The covariance seen by the beamformer is $R = \sigma^2 I + P uu^T$.

(a) Build R .

$$R = \boxed{}$$

(b) Compute the MVDR weights $w = R^{-1}s/(s^T R^{-1}s)$.

$$w = \boxed{}$$

(c) Evaluate the beamformer's response to the target, $w^T s$,
and to the interferer, $w^T u$.

$$w^T s, w^T u =$$

$$\boxed{}$$

(d) Which lab procedure did MVDR just reproduce, and what would w become if the interferer were switched off?

5. **Choosing an approach.** Each situation below uses the same 8-element PHASER.

- (a) A fixed ground emitter sits at a surveyed bearing of $+35^\circ$ from the array face and does not move. The array must keep its beam on a target at -10° . Which approach would you use, and why?

- (b) An airborne jammer of unknown and changing bearing appears during a mission. Which approach would you use, and why?

- (c) A requirement calls for a 33 dB notch on a board whose coarse 3-bit phase shifters hold a null only to about 21 dB (Lesson 27). Taking roughly 6 dB of improvement per added bit, how many phase bits are needed?

$B =$

- (d) Two jammers now illuminate the array from different bearings. State what happens when MVDR is asked to handle both on this hardware.

Documentation: _____